Questions

Bookmark this page

problem

1.0/1.0 point (graded)

In a hypothetical scenario suppose when running an A* search, if for every node the value of g(n) = 0, then which of the following search techniques would work similar to the above mentioned A* search?

- ( ) UCS
- (•) Greedy best first search
- ( ) IDS
- ( ) BFS

✔

Submit You have used 1 of 1 attempt

Show Answer

problem

1.0/1.0 point (graded)

You have designed a search algorithm, where the evaluation function is f(n) = (2-x)g(n)+xh(n). What kind of search will your designed algorithm perform if x = 1?

- ( ) Uniform cost search
- ( ) Greedy best first search
- (•) A* search
- ( ) All of the above

✔

Submit You have used 1 of 1 attempt

Show Answer

problem

1.0/1.0 point (graded)

What is the heuristic function of uniform cost search?

- ( ) f(n) not equals to g(n)
- ( ) f(n) less than g(n)
- (•) f(n) equals g(n)
- ( ) f(n) greater than g(n)

✔

Submit You have used 1 of 1 attempt

Show Answer

problem

1.0/1.0 point (graded)

If h(n) = 0, we can say that n is a:

- ( ) Source node
- (•) Goal node
- ( ) Leaf node
- ( ) None of the above

✔

Submit You have used 1 of 1 attempt

Show Answer

problem

1.0/1.0 point (graded)

If a given graph's path costs are the same, and queue is used for UCS then it will behave similar to:

- [x] BFS
- [ ] DFS
- [ ] Greedy Best First
- [ ] A*

✔

Submit You have used 1 of 1 attempt

Show Answer

problem

1.0/1.0 point (graded)

Which of the following statements about A* search are not true?

- [x] It is basically a greedy best first search.
- [x] Heuristic function plays little to no role in improving the efficiency of A-star search.
- [ ] Here, we expand nodes with the lowest f(n) cost.
- [x] We stop after we enqueue the goal node.
- [ ] Admissible heuristic never overestimates the true cost.

✔

Submit You have used 1 of 1 attempt

Show Answer

problem

0.25/1.0 point (graded)

Which of the following(s) is/are true?

- [x] Heuristic consistency ensures admissibility ✱
- [ ] Heuristic admissibility ensures consistency
- [x] For 3 nodes a, b, and c: if h1(a) = 4, h1(b) = 2, h1(c) =0 and h2(a) = 5, h2(b) = 2, h2(c) =0. Considering that there's no other node in the graph, we can say that h2 dominates h1 ✱
- [x] Greedy best first search is optimal for a tree, not for a graph
- [ ] A* combines uninformed and informed search techniques ✔

✱

Submit You have used 1 of 1 attempt

Show Answer

Answers are displayed within the problem

problem

0.5/1.0 point (graded)

Which of the following are false for greedy best first search?

- [ ] Being stuck in infinite loops
- [x] Ensures optimal solution ✱
- [ ] It uses an admissible heuristic function ✔
- [x] It is slower than A* search ✱
- [ ] None of the above

✱

Submit You have used 1 of 1 attempt

Show Answer

Answers are displayed within the problem

problem

1.0/1.0 point (graded)

Greedy best first search is:

- [x] Not complete
- [x] Not optimal
- [ ] Tends to take same amount of time as DFS
- [ ] Similar time and space complexity as DFS
- [ ] Can only be applied on weighted graph

Submit You have used 1 of 1 attempt

Show Answer

Answers are displayed within the problem

problem

1.0/1.0 point (graded)

Which of the following are true for A* search?

- [x] Consistent heuristics ensures admissibility
- [ ] Admissible heuristics ensures consistency
- [x] A* is a combination of UCS and Greedy best first search
- [ ] For inadmissible heuristic A* still returns optimal solution
- [x] A* expands less node thus it is faster than UCS

Submit You have used 1 of 1 attempt

Show Answer

Previous Next